

Towards Gamifying HCI Instruments: A Crosswalk and Equivalence Protocol for Card Sorting

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Abstract—Games and game-like artifacts have been widely studied as effective vehicles for learning, including for exposing non-experts to abstract concepts. Besides, card sorting is a foundational Human-Computer Interaction (HCI) method to obtain the user mental model, but game-based variants that seek to enrich the protocol must first demonstrate methodological fidelity. We frame this requirement as an equivalence hypothesis: at the level of the dependent variables used by researchers, a game-based card-sorting instrument should produce measurements comparable to the traditional protocol. Beyond card sorting, our goal is to establish a reusable workflow for designing research instruments that can later be applied in more complex HCI contexts, including studies where researchers have limited prior information about the participants’ context of use. We contribute a Card Sorting–Mechanics Crosswalk that maps the atomic actions of task-based card sorting onto document-inspection game mechanics while preserving the protocol’s dependent-variable structure. We instantiate the crosswalk in a 2D drag-and-drop game prototype inspired by *Papers Please*, and define a within-subject equivalence protocol for evaluating it against a traditional digital card sort. In this work in progress, data collection is complete and the primary equivalence test is ongoing, with full results reserved for a companion article. Preliminary evidence shows near-identical workload and engagement patterns across conditions, supporting the feasibility of the approach.

Index Terms—Human-Computer Interaction, User Experience, Card Sorting, Context of use, Gamification, Equivalence testing.

I. INTRODUCTION

Card sorting is a foundational method for HCI research, used to define the information architecture that supports the design of interfaces in different domains and stages of software engineering process and resulting products [1], [2]. Its established protocol has known practical limits that motivate the exploration of game-based instruments, but such instruments cannot be justified by engagement alone. If a game-based variant is to replace or extend an established HCI method, it must first show methodological fidelity: at the level of the dependent variables used by researchers, it should produce measurements comparable to the traditional protocol.

We articulate this requirement as an *equivalence hypothesis* and take a first step toward answering it for task-based card sorting. Beyond this specific method, our long-term objective is to define a reusable workflow to transform HCI research

procedures into game-based instruments that can be applied later in more complex contexts of use where difficulties arise in user research due to the lack of information and/or unavailability of users. Our contributions are:

- A *Card Sorting–Mechanics Crosswalk* that maps the atomic actions of open card sorting onto document-inspection game mechanics while preserving the dependent-variable structure of the traditional protocol.
- A within-subject equivalence protocol for testing the resulting instrument under a narrowed scope of task-based card content presented in stages of increasing ambiguity.
- A first open-source prototype, which instantiates the crosswalk as a 2D drag-and-drop document-inspection game inspired by *Papers Please* [3] and serves as the experimental vehicle of this study.

We position the framework as conservative on its equivalence claim and exploratory on its broader motivation. The present paper asks whether a game-based instrument can preserve the methodological function of card sorting while opening a path toward richer instrumentation in contexts where conventional HCI inquiry has less prior structure.

II. BACKGROUND AND MOTIVATION

Card sorting and its limits. Open card sorting is a foundational HCI research method to elicit users’ mental models of content organization, with practices and analytical tools that have matured over two decades [1], [2]. The method’s known limitations are practical rather than conceptual: protocol fatigue and drop-out reduce the quality of data on long card banks; think-aloud or retrospective elicitation of grouping rationale is expensive and noisy; and the resulting information architectures have been criticized as content-centric and unrepresentative of how users actually attempt tasks [4]. These limitations motivate ongoing methodological refinements, including modified variants and task-based formulations of the card content [2], [4], [5]. A game-based reformulation is attractive for a different reason: it can be deployed unsupervised and at scale, amenable to remote crowdsourcing, which targets the fatigue and drop-out economics of long banks, while its mechanics can capture grouping rationale in context rather than through costly think-aloud. We pursue gamification for this methodological reach, not exclusively for engagement, contingent on the instrument first measuring what the analog protocol measures.

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Games as research instruments. The use of games and game-like artifacts in research has been studied predominantly through the lens of engagement and learning, building on a tradition of gamification [6] and serious-game design and evaluation [7]. In emerging computing paradigms, game-based simulations have been documented as effective vectors for exposing non-experts to abstract concepts [8], [9]. A separate, smaller line of work treats games as instruments of measurement, vehicles through which research data are produced under controlled conditions. This instrument framing raises the question that engagement framing does not: under what conditions does a game-based instrument produce measurements equivalent to the traditional research protocol it stands for? We are not aware of work that has explicitly answered this question for card sorting.

Transferable HCI-UX instrumentation. Our longer-term goal is a workflow for game-based HCI instruments deployable in contexts of use that are not yet well characterized—where ISO 9241-210’s call for an explicit understanding of context [10] is hardest to satisfy. Such an instrument must first preserve the function of the method it re-instruments; we take that first step in card sorting, in the familiar domain of education.

III. THE CARD SORTING–MECHANICS CROSSWALK

We use the term *crosswalk* for a structured, action-by-action mapping between two systems—here, from card-sorting actions to game mechanics. Game-based instrumentation of an established HCI research method is a methodologically responsible move only if the new instrument preserves the function of the original. We articulate this requirement as an *equivalence hypothesis*: at the level of the dependent variables that researchers use to inform interface design, *a game-based card sorting instrument should produce measurements indistinguishable from those of the traditional protocol*. As a first exploratory attempt, we evaluate the hypothesis within a deliberately narrowed scope: task-based card content presented in stages of increasing ambiguity.

A. Scope of this first attempt

Task-based content. Card sorting (CS) is practiced in two main variants: *label-based*, in which cards bear navigation or category labels (“nouns”), and *task-based*, in which cards bear user goals or actions (“verbs”) [1], [2]. The label-based variant has been criticized as inherently content-centric and prone to producing information architectures that perform poorly when users attempt actual tasks [4]. We adopted task-based content for this first attempt for two reasons. First, it shifts the participant’s cognitive activity from taxonomic judgment to functional grouping, which is the activity that is actually relevant to interfaces whose value is realized through user work [2], [4]. Second, task-based content surfaces inter-participant variability more strongly than label-based content, which is essential to make a positive equivalence finding non-trivial.

TABLE I
CARD SORTING–MECHANICS CROSSWALK PROPOSAL.

CS action	Game mechanic
<i>Primary-DV (co-occurrence matrix)</i>	
Inspect card	Pick up document from queue, read details
Place in group	File document in folder
Move between groups	Re-file document into a different folder
Create new group	Open a new folder
Name group	Stamp folder with a label
Rename group	Replace folder label
Merge groups	Consolidate n folders into one
Split group	Re-file a folder’s contents into new folders
<i>Secondary-DV (workload, time, drop-out)</i>	
Discard card	Reject document into the discard tray
End-of-task review	End-of-shift folder review

Staged-by-ambiguity presentation. The default open CS protocol presents the full card pile to the participant in a single pass. Iterative variants exist – most notably the modified Delphi approach, in which participants revisit categorizations across rounds with structured feedback [5] – demonstrating that non-single-pass CS is not without methodological precedent. In the present study, we depart from the single-pass default along a different axis: cards are presented in stages of increasing item ambiguity, mirroring the staged-difficulty pacing common to game-design practice [11], [12]. We adopt this purely as a scope choice consistent with the staged structure of the game-based instrument.

B. Construction principle

We classified each atomic CS action by its role in the dependent-variable (DV) structure of the protocol. Actions that contribute directly to the primary co-occurrence matrix (the substrate on which Jaccard agreement is computed) are designated *primary-DV actions* and must be preserved one-for-one in the game. Actions that contribute to secondary measures (workload, engagement, decision time, drop-out) are *secondary-DV actions* and admit limited transformation provided their methodological function survives. We then matched each action with the closest mechanic in the document-inspection family popularized by *Papers Please* from Lucas Pope [3], verifying for each that the mechanic preserves the action’s methodological function rather than its surface gesture. Three CS actions admit *enhancements*: features that the game format affords beyond what the analog protocol can practically support. The mapping is summarized in Table I.

IV. IMPLEMENTATION

We instantiate the crosswalk in *Cards Please*, an open-source Unity 6 WebGL build that we use as the experimental vehicle for the present study¹. We confine the description below to the design decisions that materially affect the equivalence claim and that distinguish the instantiation from a generic re-skin.

¹The implementation, with bilingual localization (ES/EN), is available in a GitHub repository: <https://anonymous.4open.science/r/Cards-Please-v012>

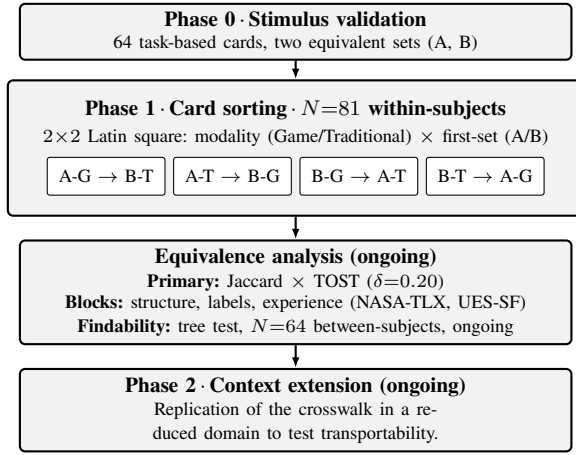


Fig. 1. Pre-registered protocol overview.

Diegetic frame. The narrative wrapper situates the participant as a new intern in a university student-services office, processing incoming task requests over a five-day work week. Levels 2–5 deliver the experimental cards in a 6–8–10–8 distribution of increasing item ambiguity, realizing the staged-by-ambiguity scope condition of §III-A. Level 1 is a tutorial situated in an unrelated domain chosen to avoid categorical priming on the experimental cards.

Core loop. Each card arrives as an incoming document in a queue. The intern picks it up, reads the task it describes, and drags it onto the desk, either into an existing folder or into a new one, which can be labelled or renamed. A level ends when its queue is empty; the five-day frame chains the levels into one within-subjects session.

Action-based pacing. The arrival of new cards is governed by a queue advanced by participant actions, not by wall-clock time. This removes temporal pressure as a confounding factor and keeps decision time interpretable as a process variable in the secondary-DV space rather than as a reaction-time score under stress.

Logging The log captures (i) raw events with placements, moves, discards, zone creation, centroids, label assignment with timestamps, and (ii) an end-of-session snapshot enriched with the final folder structure and label set.

V. PRE-REGISTERED EQUIVALENCE PROTOCOL

We evaluate the equivalence hypothesis through a pre-registered within-subject study comparing an established card-sorting protocol with the game-based instantiation described in §IV. The protocol was specified prior to data collection in Phase 1; its core design and the operational criteria that define the equivalence at the level of dependent variables are summarized below and illustrated in Fig. 1.

Design. Participants ($N=81$ enrolled students of the Computer Science degree at the ETSI Informáticos, UPM) completed two conditions: a game-based condition using *Cards Please* (§IV) and an established card-sorting condition (*traditional*) carried out on a collaborative digital whiteboard².

Each participant processed both sets (Set A and Set B, 32 cards each) drawn from the master bank described in Phase 0 below. Participants were assigned to one of four sequences in a 2×2 Latin-square design that crosses the first-modality (Game/Traditional) with the first-set (A/B). The master bank covers the student and instructor roles in a learning-management system³, stratified by category. With $N=81$ and a contrast between subjects, the design substantially exceeds the card-sorting conventions [13] and provides $\geq 80\%$ power for the TOST procedure at $\delta=0.20$ in plausible SD scenarios.

Primary measure and equivalence test. The primary dependent variable is the Jaccard agreement between the participant-by-participant co-occurrence matrices produced under the two conditions. We will test equivalence using the two one-sided tests (TOST) procedure [14], [15].

Four equivalence blocks. Beyond the primary Jaccard test, we examine equivalence across four complementary blocks that together characterize the methodological fidelity of the game-based instrument: (i) *structure*—number and size distribution of groups; (ii) *labels*—semantic similarity of group names; (iii) *findability*—first-click success in a tree test between-subjects ($N=64$, separate participants) over information architectures derived from each condition; and (iv) *experience*—NASA-TLX workload and UES-SF engagement scores [16]–[18].

Secondary measures and exploratory analyses. Decision time per card and drop-out rate are collected as secondary measures. Analyses that involve enhanced mechanics are exploratory and do not affect the primary equivalence claim.

Phase 0: stimulus validation. The card bank was constructed by the authors via systematic scraping of the target platform’s task surface followed by manual refinement, with an a priori classification of each card on difficulty and category ambiguity. Three HCI faculty then completed an independent review that combined (i) an open sort of both Set A and Set B (median session time 21 min), (ii) per-card clarity, representativeness, and category ambiguity ratings – blind to the authors’ a priori classification – and (iii) written feedback on individual cards. Cards flagged as ambiguous in either pass received targeted edits before Phase 1. The reviewers also selected the operational domain by anonymous vote.

Phase 1 status. Card-sorting data collection was completed prior to submission; analysis is in progress, and empirical results are reserved for a follow-up article. The findability tree test is ongoing.

Phase 2 status. A between-subjects extension applies the same crosswalk in a reduced computing context, addressing the transportability hypothesis articulated in §VI. This phase is currently in preparation and serves as an empirical bridge to a domain where this technique has not been applied yet.

VI. DISCUSSION AND ONGOING WORK

Preliminary subjective evidence. Across all 81 participants, both conditions produced near-identical responses on

²Miro platform: <https://miro.com>

³Inspired by the Moodle platform: <https://moodle.com>

TABLE II
CARDS PLEASE VS. TRADITIONAL CS: SUBJECTIVE MEASURES.

Measure	CP $M(SD)$	Tr $M(SD)$	Δ [95% CI]
NASA-TLX RTLX	3.06 (1.08)	3.11 (1.08)	-0.06 [-0.32, +0.21]
UES-SF Total (1-5)	3.66 (0.66)	3.67 (0.65)	-0.01 [-0.19, +0.17]
Focused Attention	3.83 (0.80)	3.88 (0.71)	-0.05 [-0.25, +0.15]
Perceived Usability	3.75 (0.90)	3.69 (0.98)	+0.06 [-0.22, +0.34]
Aesthetic Appeal	3.40 (0.96)	3.43 (1.06)	-0.03 [-0.36, +0.30]
Reward	3.67 (0.79)	3.70 (0.87)	-0.03 [-0.27, +0.21]

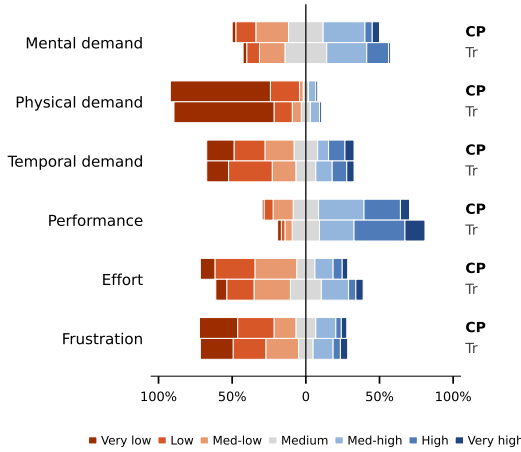


Fig. 2. Cards Please vs Traditional CS: NASA-TLX distribution.

both workload and engagement tests (Table II). The item-level response patterns superpose between conditions (Fig. 2).

However, the two equivalent designs differ in how they distribute residual variance: TLX- Δ varied between the four counterbalanced groups with a first-modality effect ($p < .001$) and a first-set effect ($p < .001$), producing a range per-group from -1.30 to $+0.63$; UES-SF showed neither effect ($p > .85$ for both). The Latin-square design absorbs this heterogeneity at the aggregate level (which is what counterbalancing is for), and the cleanliness of the UES-SF data itself is a positive methodological signal. That said, the descriptive pattern is encouraging for the equivalence hypothesis.

For the Games research community. The crosswalk treats game mechanics as instruments of measurement rather than vehicles of engagement: a good mechanic for a research-purpose game is one that preserves the methodological function of the analog action it stands for—an instrumentation analogue of meaningful play [19]. If equivalence is supported, it would open up an opportunity to transfer the same instrumentation workflow to other HCI research methods whose outputs depend on recoverable participant actions.

Transportability. The same crosswalk is being applied in a more specialized domain with a smaller expert pool, as a first test of whether the instrument transports to contexts of use that are less characterized than a learning-management system. We situate the present validation in a familiar domain, with real non-expert users, so that domain unfamiliarity does not confound the equivalence test before the instrument is moved.

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REFERENCES

- [1] D. Spencer, *Card Sorting: Designing Usable Categories*. Brooklyn, NY: Rosenfeld Media, 2009.
- [2] J. R. Wood and L. E. Wood, "Card sorting: Current practices and beyond," *Journal of Usability Studies*, vol. 4, no. 1, pp. 1–6, 2008.
- [3] L. Pope, "Papers, please," 2013. [Online]. Available: <https://papersplease>
- [4] C. Katsanos, V. Christoforidis, and C. Demertzi, "Task-based open card sorting: Towards a new method to produce usable information architectures," in *Human Interface and the Management of Information (HCII 2023)*, ser. Lecture Notes in Computer Science, vol. 14015. Springer, 2023, pp. 68–80.
- [5] C. L. Paul, "A modified Delphi approach to a new card sorting methodology," *Journal of Usability Studies*, vol. 4, no. 1, pp. 7–30, 2008.
- [6] S. Deterding, D. Dixon, R. Khaled, and L. Nacke, "From game design elements to gamification: defining 'gamification'," in *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments*, ser. MindTrek '11. New York, NY, USA: Association for Computing Machinery, 2011, p. 9–15. [Online]. Available: <https://doi.org/10.1145/2181037.2181040>
- [7] I. Mayer, G. Bekebrede, C. Harteveld, H. Warmelink, Q. Zhou, T. van Ruijven, J. Lo, R. Kortmann, and I. Wenzler, "The research and evaluation of serious games: Toward a comprehensive methodology," *British Journal of Educational Technology*, vol. 45, no. 3, pp. 502–527, 2014. [Online]. Available: <https://bera-journals.onlinelibrary.wiley.com/doi/abs/10.1111/bjet.12067>
- [8] P. Migdał, K. Jankiewicz, P. Grabarz, C. Decaroli, and P. Cochin, "Visualizing quantum mechanics in an interactive simulation – Virtual Lab by Quantum Flytrap," *Optical Engineering*, vol. 61, no. 8, p. 29, 2022.
- [9] A. Zable, L. Hollenberg, E. Velloso, and J. Goncalves, "Investigating immersive virtual reality as an educational tool for quantum computing," in *Proc. 26th ACM Symp. Virtual Reality Software and Technology (VRST)*, 2020.
- [10] International Organization for Standardization, "ISO 9241-210:2019 ergonomics of human-system interaction — Part 210: Human-centred design for interactive systems," 2019.
- [11] J. Schell, *The Art of Game Design: A Book of Lenses*. Morgan Kaufmann, 2008.
- [12] C. W. Totten, *An Architectural Approach to Level Design*. CRC Press, 2014.
- [13] T. Tullis and L. E. Wood, "How many users are enough for a card-sorting study?" in *Proc. Usability Professionals' Association (UPA) Annual Meeting*, Minneapolis, MN, 2004.
- [14] D. Lakens, "Equivalence tests: A practical primer for t tests, correlations, and meta-analyses," *Social Psychological and Personality Science*, vol. 8, no. 4, pp. 355–362, 2017.
- [15] D. Lakens, A. M. Scheel, and P. M. Isager, "Equivalence testing for psychological research: A tutorial," *Advances in Methods and Practices in Psychological Science*, vol. 1, no. 2, pp. 259–269, 2018.
- [16] S. G. Hart and L. E. Staveland, "Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research," in *Advances in Psychology*, 1988, vol. 52, pp. 139–183.
- [17] S. G. Hart, "NASA-Task Load Index (NASA-TLX); 20 years later," in *Proc. Human Factors and Ergonomics Society Annual Meeting*, vol. 50, no. 9, 2006, pp. 904–908.
- [18] H. L. O'Brien, P. Cairns, and M. Hall, "A practical approach to measuring user engagement with the refined User Engagement Scale (UES) and new UES Short Form," *Int. J. Human-Computer Studies*, vol. 112, pp. 28–39, 2018.
- [19] K. Salen and E. Zimmerman, *Rules of Play: Game Design Fundamentals*. MIT Press, 2003.